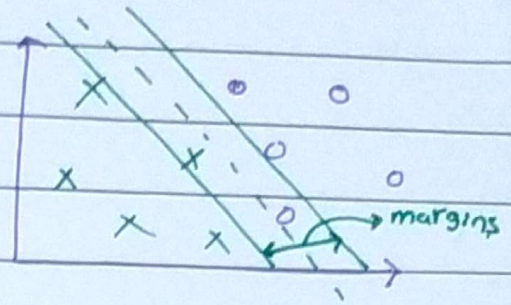
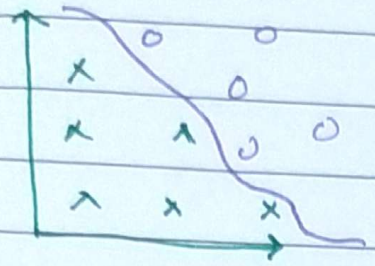
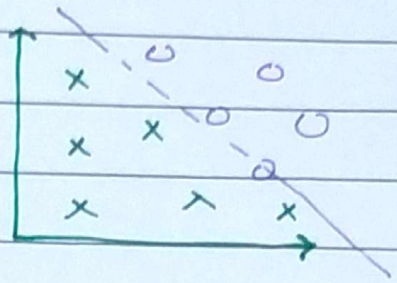
SVM: To separate 2 classes & minimize the margins between nearby data points



→ For n dimentions data is separated by hyper plane



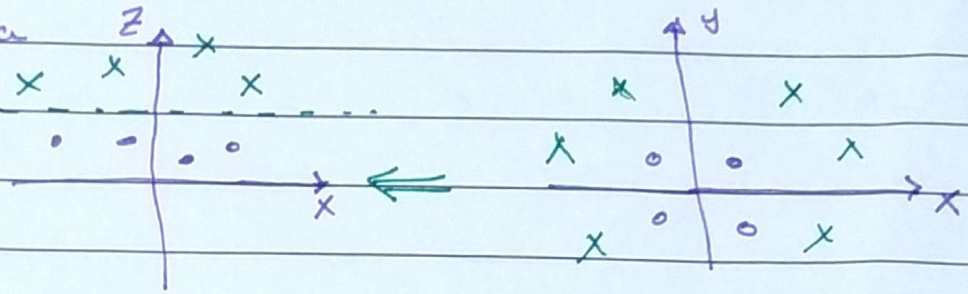


High Regulization

low Regulization

→ If our features overlapped we can add another dimention To separate the data

$Z = x^2 + y^2$ "kernel"

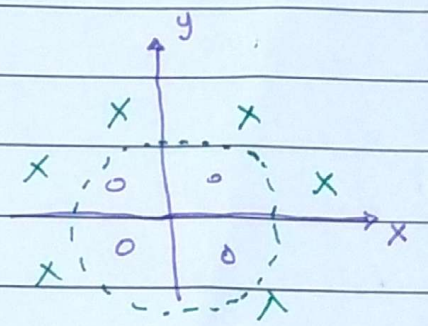


① Building method →

② Circular method

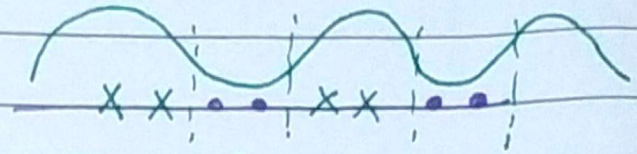
→ separate them using a circle

→ Both circular & building Tricks are equivalent



②

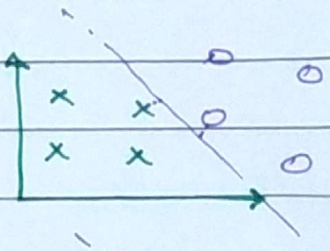
③ RBF Method "Trik" [Radial bases Function] kernel



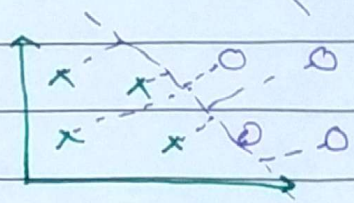
Gamma γ
- → large Gamma → narrow curve
- → Small Gamma → wide curve

$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \rightarrow \gamma = \frac{1}{2\sigma^2}$$

Normal distribution



High Gamma ⇒ consider near by Points only



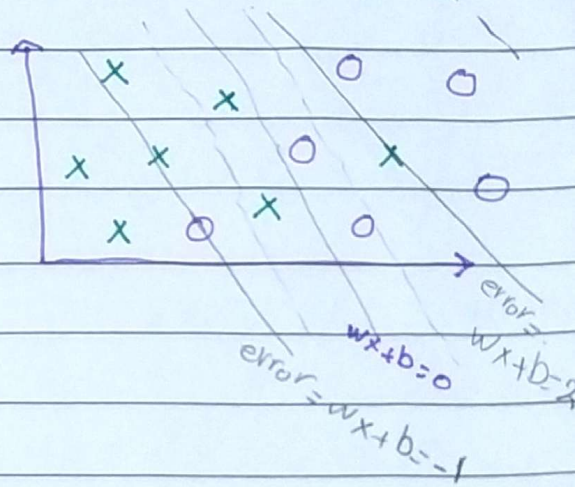
low Gamma → consider both near and far Points

ERROR :-

→ Clasification error النقط اللي مش فيها

classification error = 1 + 2

→ Margin error



margin = $\frac{2}{|w|}$ margin error = $|w|^2$.

large Margin → small error

small margin → large error

Total error = C * classification error + margin error

C Prameter
- → large C → focus on classifying
- → small C → focus on large margin